AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

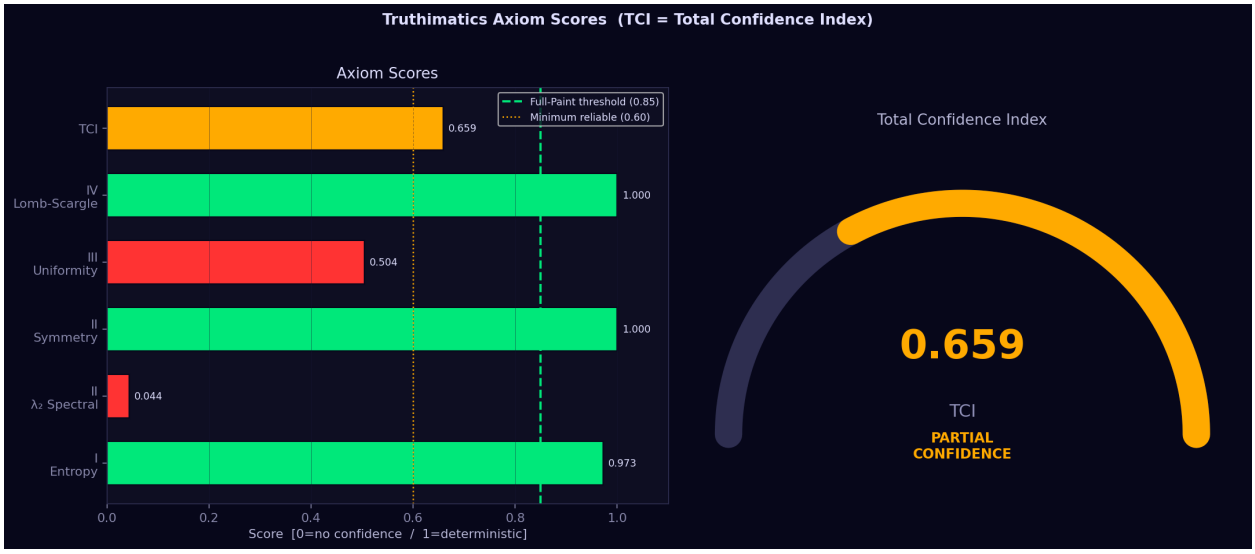
Specimen ID	AXIOM_ECE346F03F
Taxa	Struthiomimus
Clade	Theropod
PBDB Records	18
Mass (regression)	342.8 kg (Campione & Evans 2012)
Mass (volumetric)	793.5 kg (Silhouette integral, ρ=1060.0 kg/m³)
Body Volume	0.934 m³
Validity Score	1.0000 / 1.0000
TCI	0.6586 (partial)
Axiom I Entropy	0.9726
Axiom II λ■	0.1763 (symmetry 100%)
Axiom III Uniformity	0.5043
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	56.0% Measured / 44.0% Inferred
Elements	25
CoG (m)	[0.202, 0.405, 0.0]
Generated	2026-04-30 23:54 UTC

Ratio Violations

• metatarsal/tibia=0.689 outside [0.28,0.65]

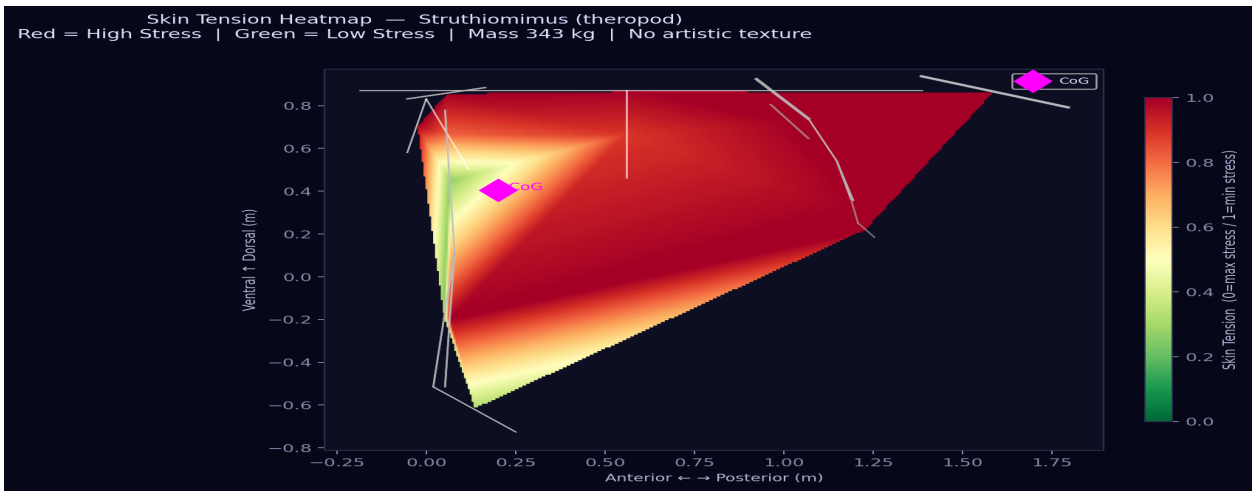
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



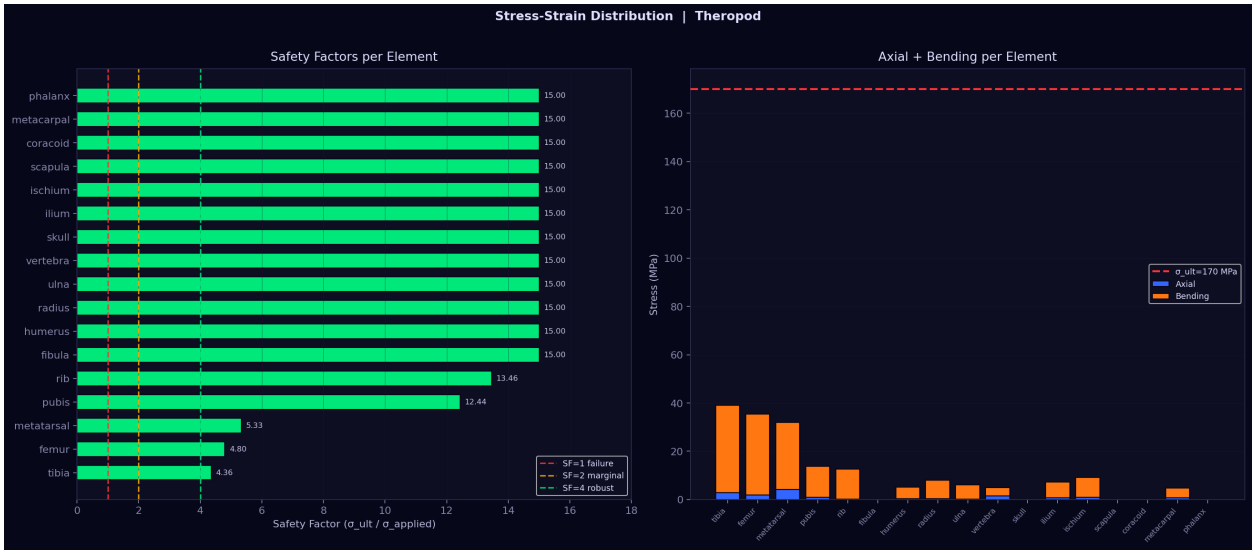
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

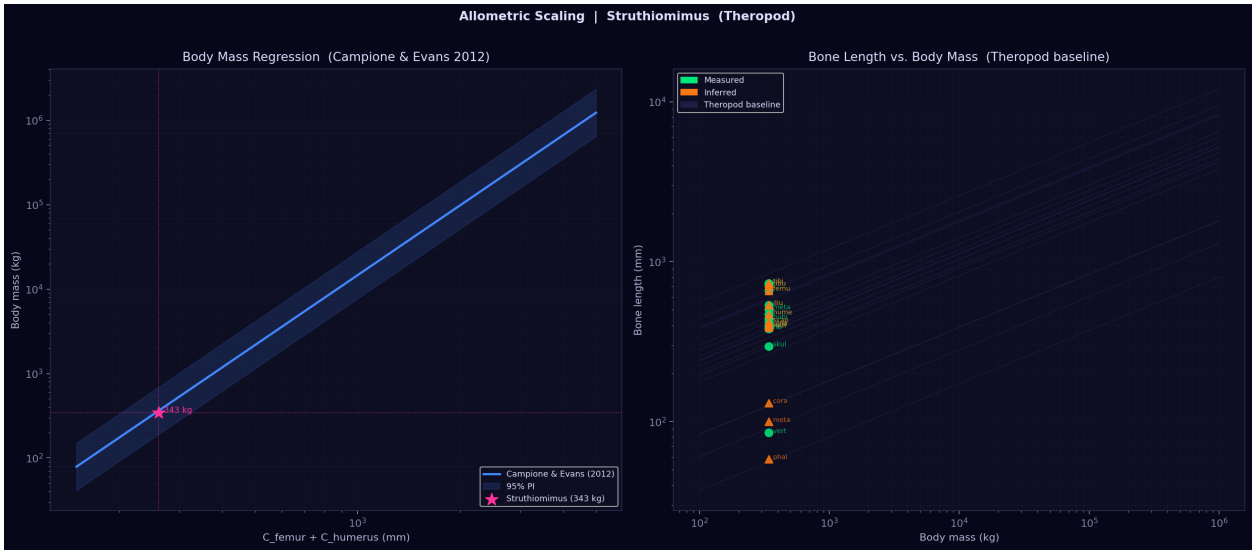


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.018	0.040	0.058	15.000	OK
femur	1.822	33.627	35.449	4.796	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.274	4.920	5.194	15.000	OK
ilium	0.810	6.461	7.271	15.000	OK
ischium	0.833	8.305	9.138	15.000	OK
metacarpal	0.737	3.934	4.671	15.000	OK
metatarsal	4.099	27.788	31.887	5.331	OK
phalanx	0.000	0.000	0.000	15.000	OK
pubis	1.017	12.651	13.668	12.437	OK
radius	0.294	7.652	7.946	15.000	OK
rib	0.243	12.391	12.634	13.456	OK
scapula	0.008	0.040	0.049	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	2.907	36.129	39.036	4.355	OK
ulna	0.243	5.927	6.170	15.000	OK
vertebra	1.552	3.406	4.959	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	1.898 m	Femur + tibia + metatarsal
Estimated Speed	12.9 km/h (3.59 m/s)	Gait: Walk | Fr=0.693
Stride Length	3.9 m	Alexander (1976) / Thulborn (1990)
Bite Force	224.7 N (22.9 kgf)	Calibrated theropod regression (Bates & Falkingham 2012)
Skull Length	295.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.2285	Circumference / length (dimensionless)
Limb Strength Index	3e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	5.75 m	Allometric from femur length
Est. Neck Length	0.81 m	Allometric from femur length
Est. Tail Length	2.19 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	131.0	80.2	337.23	421.0	Constraint	0.40	15.00
femur	660.0	48.0	609.10	760.3	Measured	0.88	4.80
femur_right	660.0	48.0	609.10	760.3	Symmetry	0.84	4.80
fibula	710.0	14.0	39.35	41.3	Measured	0.88	15.00
fibula_right	710.0	14.0	39.35	41.3	Symmetry	0.84	15.00
humerus	468.0	35.0	229.64	286.7	Measured	0.88	15.00
humerus_right	468.0	35.0	229.64	286.7	Symmetry	0.84	15.00
ilium	535.0	72.0	1110.91	1386.7	Measured	0.88	15.00
ilium_right	535.0	72.0	1110.91	1386.7	Symmetry	0.84	15.00
ischium	390.0	42.0	275.56	344.0	Measured	0.88	15.00
metacarpal	99.9	20.7	6.38	5.2	Constraint	0.65	15.00
metatarsal	505.0	32.0	207.13	258.6	Measured	0.88	5.33
phalanx	58.2	16.7	1.96	1.5	Constraint	0.40	15.00
pubis	440.0	38.0	254.50	317.7	Measured	0.88	12.44
radius	388.0	20.0	62.17	77.6	Measured	0.88	15.00
radius_right	388.0	20.0	62.17	77.6	Symmetry	0.84	15.00
rib	380.0	12.0	21.92	27.4	Measured	0.88	13.46
scapula	420.0	118.0	2342.47	2924.1	Measured	0.88	15.00
scapula_right	420.0	118.0	2342.47	2924.1	Symmetry	0.84	15.00
skull	295.0	125.0	1846.30	2304.7	Measured	0.88	15.00
tibia	733.0	38.0	423.97	529.2	Measured	0.88	4.36
tibia_right	733.0	38.0	423.97	529.2	Symmetry	0.84	4.36
ulna	400.0	22.0	77.55	96.8	Measured	0.88	15.00
ulna_right	400.0	22.0	77.55	96.8	Symmetry	0.84	15.00
vertebra	85.0	52.0	92.06	114.9	Measured	0.88	15.00